

MA 214 Week 11: Lab 3

Bayes' Rule

1. **Your name:** _____

BU ID: _____

2. **Partner's name:** _____

BU ID: _____

Big idea: Bayes' Rule updates beliefs by combining prior probabilities with the likelihood of the observed data. **Do not use GenAI tools for this lab.** The point is to practice your own analysis work, coding skills, and statistical reasoning.

Files used: coin files in `coin_data_in_lab/`, `lab3_post_data.csv`, and `lab-starter.R`

Skills practiced: `dbinom()`, vectorized probability calculations, posterior normalization, and sequential updating.

Your mission: Use coin-flip data to infer which of five possible coin types the dealer gave you.

Icebreaker

Before opening R: introduce yourself to your partner and answer the warm-up question below.

Warm-up response: If a coin lands heads 8 times in 10 flips, what would you believe about the coin before doing any formal calculation?

Getting Started

Open `lab-starter.R`. Choose or use your assigned coin file, then load the data and count the number of heads.

```
library(dplyr)
library(ggplot2)

my_coin_file <- file.path("coin_data_in_lab", "coin_01.csv")
coin_df <- read.csv(my_coin_file)

p_js <- c(0.15, 0.30, 0.50, 0.70, 0.90)
coin_labels <- c("A", "B", "C", "D", "E")

n_flips <- nrow(coin_df)
n_heads <- sum(coin_df$result)
```

HUDDLE UP

Before calculating anything, which coin type seems most plausible from the sample proportion of heads? What makes you unsure?

Question 1. Explore the Coin Data

Record the possible coin types, the observed number of flips, and the observed number of heads.

```
n_flips  
n_heads  
n_heads / n_flips
```

Variable or data set	Type or role	What it means
p_js		
coin.df\$result		
prior_probs		

Write 2-3 sentences describing what the data suggest before applying Bayes' Rule.

Question 2. Make a Prediction

Before computing the posterior, predict which coin type will have the highest posterior probability.

PREDICTION TIME

Use the sample proportion of heads to make a prediction. Be specific enough that the posterior can support or challenge your answer.

```
prior_probs <- rep(1 / 5, 5)  
  
likelihood <- dbinom(n_heads, size = n_flips, prob = p_js)  
numerator <- likelihood * prior_probs  
posterior_batch <- numerator / sum(numerator)
```

What did your team predict? What evidence did you check first?

Question 3. Compute the Posterior

Compare the batch posterior with the posterior from sequential updating.

```
post_mat <- matrix(NA, nrow = n_flips + 1, ncol = 5)
post_mat[1, ] <- prior_probs

for (i in 1:n_flips) {
  lk <- ifelse(coin_df$result[i] == 1, p_js, 1 - p_js)
  unnorm <- post_mat[i, ] * lk
  post_mat[i + 1, ] <- unnorm / sum(unnorm)
}

posterior_seq <- post_mat[n_flips + 1, ]
```

HUDDLE UP

Do the batch and sequential posterior probabilities match? What does that tell you about Bayesian updating?

CLAIM, EVIDENCE, CONTEXT

The most likely coin type is _____.

My evidence is _____.

This matters because _____.

Question 4. Interpret in Context

Use posterior probabilities to make a decision about the coin type.

Question	Your answer
What comparison did you make?	
What result is strongest?	
What limitation or caution should readers know?	

DEBATE CORNER

If you and your partner disagree about the final coin type, write both arguments first. Then decide what claim the evidence actually supports.

Question 5. Close the Lab

Submit your completed `lab-starter.R`. Your submitted file should define the required posterior objects listed in the starter file.

Final response: state your coin-type claim, cite one posterior probability, and explain the context.

Optional Challenge

Plot the posterior probability of each coin type after every flip.

```
# Optional challenge starter code.  
matplot(post_mat, type = "l", lty = 1)
```